

FACE RECOGNITION PRO

Automated Smart Attendance Using Computer Vision

A real-time, camera-based attendance management system built to replace manual roll-calls and stop proxy attendance.

Python

OpenCV

NumPy

CSV

DateTime



Project Overview



Face Recognition Pro is an AI-powered attendance management system that replaces manual roll calls with real-time, camera-based identification. The system registers by capturing a student's facial data, trains a lightweight recognition model with OpenCV, and continuously scans incoming webcam frames for a match. The moment a face is confirmed, the student's name, date, and time are logged automatically — removing paperwork, human error, and the chance of marking attendance for someone who isn't present (proxy attendance).

40

reference images captured per student

Real-Time

detection and recognition via webcam

Auto-CSV

attendance export, Excel-compatible

Problem Statement



Manual attendance is time-consuming

Calling out names or passing a sheet around eats into class time every single session.



Proxy attendance is common

Students mark or answer for absent classmates, making manual registers unreliable.



Large classrooms are hard to manage

Bigger batches make manual tracking slower, messier, and more error-prone.



THE SOLUTION

**Automate attendance using face
recognition**

A single camera and a trained model do what a roll call can't:
verify identity instantly, every time.

Objectives



Automate the attendance process

Remove manual roll-calls entirely by letting the camera confirm who is present.



Reduce manual effort & errors

Cut down on data-entry mistakes and the time staff spend on registers.



Improve accuracy with face recognition

Use verified facial identity instead of self-reported or proxy marking.



Store attendance data digitally

Keep clean, timestamped CSV records ready for reporting and analysis.

Technologies Used



Python

Core programming language for the entire pipeline



OpenCV (cv , cv2)

Face detection, image processing & recognition



NumPy

Fast array operations on image & pixel data



CSV

Lightweight, Excel-compatible attendance storage



DateTime

Accurate time stamping of every attendance entry

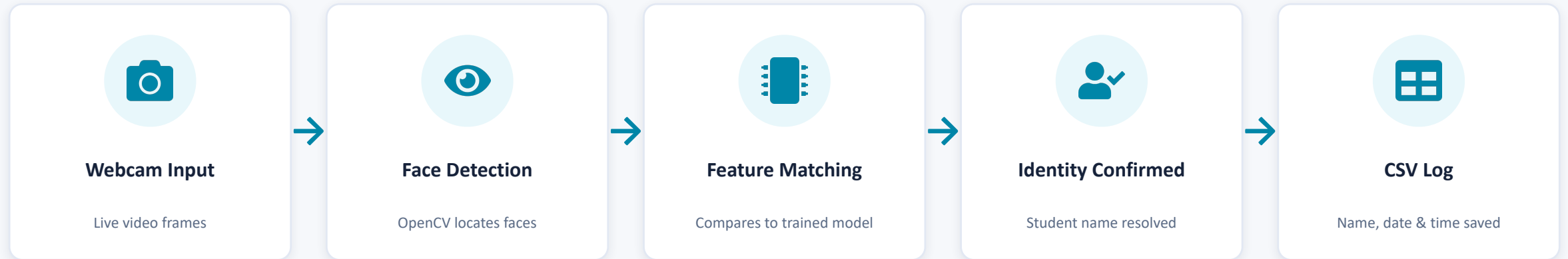


OS & JSON

File handling and structured configuration data

System Architecture

How data flows from a webcam frame to a saved attendance record



Two phases keep the system simple to run and easy to extend:

Registration phase — captures 40 face images per student and trains a recognition model from that dataset. | Recognition phase — runs continuously on the live webcam feed, matches each detected face against the trained model, and writes a timestamped record the instant a match is confirmed.

System Workflow

1

Capture student images

40 images collected per student through the webcam

2

Train the recognition model

Captured images are processed into a trained data file

3

Open webcam for detection

Live feed starts scanning for faces in real time

4

Recognise faces

Detected faces are matched against the stored dataset

5

Mark attendance

A confirmed match is stamped with date & time

6

Save records

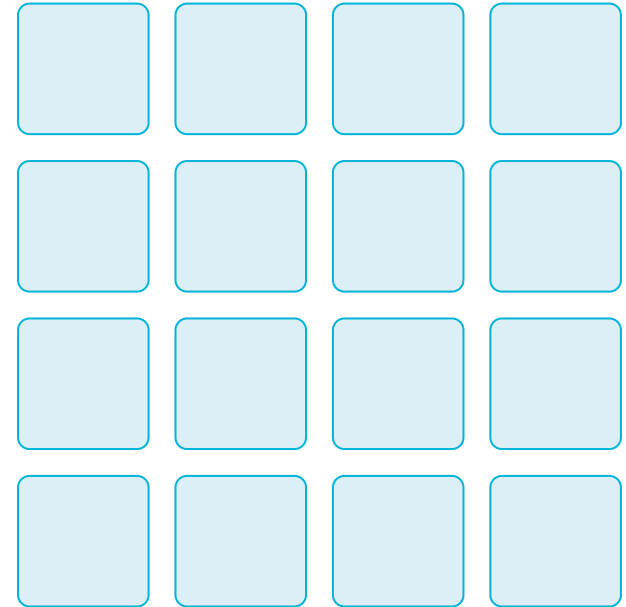
The entry is appended to the attendance CSV file

Image Collection Module



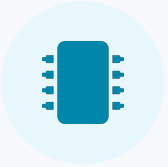
- The webcam captures roughly 40 images of each student during registration, covering slight variations in angle, distance, and expression.
- Every captured frame is saved into a dedicated per-student folder inside the dataset directory, keeping the training data organised.
- A larger, more varied image set directly improves recognition accuracy and makes the model more resilient to lighting changes.

`dataset/student_name/`



40 images / student

Model Training Module



- Every collected image is processed with OpenCV: converted to grayscale and cropped to the detected face region.
- The processed dataset is fed into an OpenCV face-recognition trainer, which learns the distinguishing patterns for each student.
- The result is a single trained model file that the recognition module loads at runtime to identify students instantly.

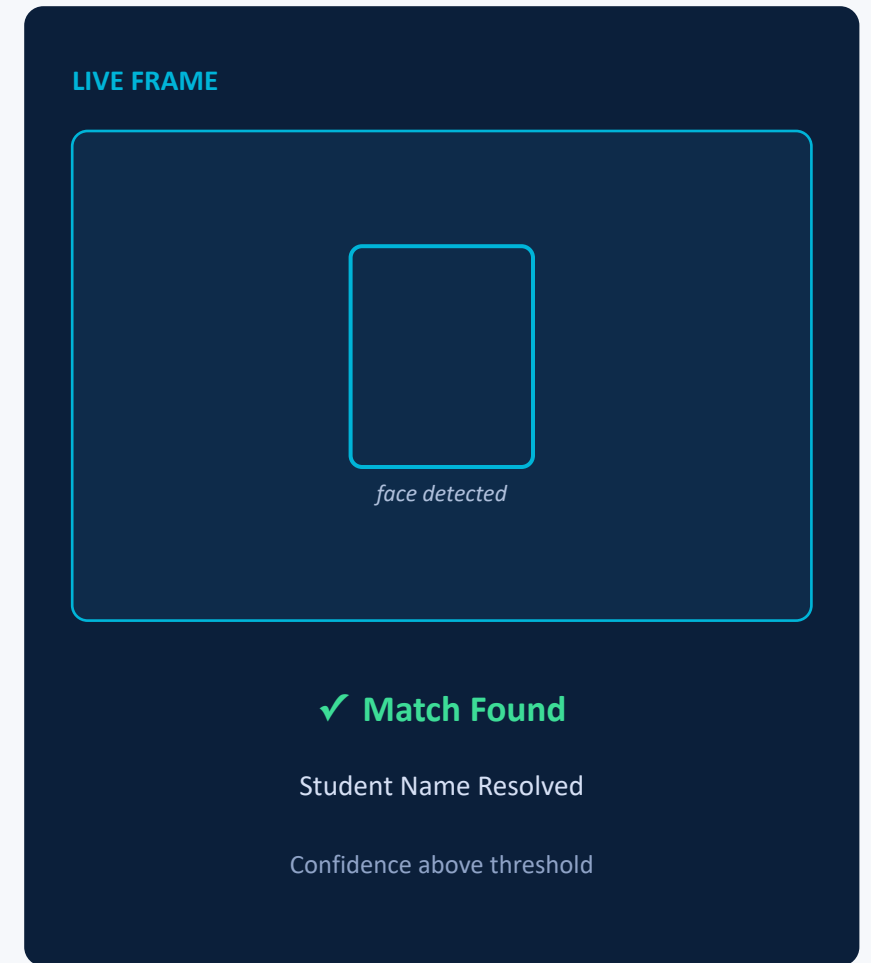


Each student's face data is stored under a unique numeric ID, linking the trained model back to their name for lookup at recognition time.

Face Recognition Module



- The webcam streams a continuous live feed once the recognition module starts.
- OpenCV detects any face present in each frame in real time.
- The detected face is compared against the trained dataset to find the closest match.
- When the match confidence clears the threshold, the corresponding student is identified by name.



Attendance Logging



Student name identified
Resolved from the trained recognition model



Current date recorded
Captured automatically at the moment of the match



Time captured automatically
Logged down to the second for accuracy

Every confirmed match is appended as a new row — no manual step required.

attendance.csv

Name	Date	Time
Aditi Sharma	2026-08-01	09:02:14
Rohan Mehta	2026-08-01	09:02:41
Kavya Nair	2026-08-01	09:03:05
Sameer Iqbal	2026-08-01	09:03:52
Kavya Nair	2026-08-02	09:05:36

Key Features



Real-time face detection

Faces are located and tracked live from the webcam feed.



Automatic attendance marking

No button presses - Just running 2 program once .



Multiple student recognition

Several students can be identified in the same session.



Fast processing

Recognises 30+ students per minute under normal conditions.



CSV data export

Records save directly in a csv format.

Advantages



Saves time

No more roll-calls eating into class or lecture time.



Eliminates proxy attendance

A face match can't be answered on someone else's behalf.



Fully automated system

Runs on its own once registration and training are complete.



Easy data management

Attendance lives in a clean, searchable CSV file.



Scalable for large classrooms

Handles growing batch sizes without extra manual effort.

Limitations

Honest constraints to plan around before deployment



Requires good lighting

Low light or strong backlight can reduce detection accuracy.



Depends on dataset quality

Fewer or poor-quality registration images weaken recognition.



Camera quality matters

Low-resolution webcams make faces harder to match reliably.

Future Enhancements



Cloud database integration

Move attendance storage from local CSV to a hosted database for multi-device access.



Mobile app support

Let teachers and admins view attendance records from a phone.



Face mask detection

Add mask-aware recognition so accuracy holds up in health-conscious settings.



College ERP integration

Sync attendance directly into existing academic management systems.

Conclusion

This project successfully demonstrates how face recognition can automate attendance systems efficiently. By pairing OpenCV's detection pipeline with a simple CSV-based log, it improves accuracy, saves classroom time, and offers a scalable foundation that can grow from a single classroom to an entire institution.



Time Saved

No manual roll-calls



Accuracy

Verified facial identity



Scalable

Ready for larger deployments





Thank You

Built with Python, OpenCV & NumPy — open for review, feedback, and contributions.

Python

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NumPy

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